

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. On HBV serology, what pattern (HBsAg / anti-HBc / anti-HBs) indicates acute infection vs chronic infection vs vaccine immunity?
2. What is the defining histologic feature that separates chronic from acute viral hepatitis?
3. Which two autoantibody profiles distinguish type 1 from type 2 autoimmune hepatitis?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Hepatitis — the frame

What	A general term for liver parenchymal inflammation .
Causes	Infective (viral, bacterial, parasitic/helminthic) · drugs · toxins · autoimmune · metabolic.
Hepatotropic viruses	HAV, HBV, HCV, HDV, HEV (other viruses e.g. CMV, EBV can also involve the liver).

Clinicopathologic syndromes of viral hepatitis

Syndrome	Features
Acute asymptomatic infection	With recovery.
Acute symptomatic infection	Recovery; signs/symptoms <6 months ; any hepatitis virus. Course: incubation → pre-icteric (malaise, weakness, RUQ pain, nausea, anorexia) → icteric (jaundice, dark urine) → convalescence.
Acute liver failure	
Chronic hepatitis	Symptomatic/biochemical/serologic evidence of continuing or relapsing disease for >6 months .
Carrier state	Harbors and can transmit the virus, no symptoms : either no liver disease, or asymptomatic non-progressive liver damage.

Comparing the hepatitis viruses

Feature	HAV	HBV	HCV	HDV	HEV
Common name	"Infectious"	"Serum"	"Post-transfusion"/ non-A, non-B	"Delta"	"Enteric"
Virus	Hepatovirus, nonenveloped capsid RNA	Hepadnavirus, enveloped DNA	Flavivirus, enveloped RNA	Defective enveloped circular RNA	Hepevirus, nonenveloped capsid RNA
Transmission	Fecal-oral	Parenteral, sexual, perinatal	Parenteral, sexual	Parenteral, sexual	Fecal-oral
Severity	Mild	Occasionally severe	Usually subclinical	Co-infection: occ. severe · superinfection: often severe	Mild (normal); severe in pregnancy
Chronicity/ carrier	No	Yes	Yes (high)	Yes	No
Clinical diseases	Acute hepatitis	Acute, chronic, cirrhosis, HCC	Chronic, cirrhosis, HCC	Acute, chronic, cirrhosis, HCC	Acute hepatitis
Diagnosis	Anti-HAV IgM	HBsAg, HBeAg, anti-HBc IgM	Anti-HCV (EIA)	Anti-HDV ELISA	Not routinely available
Prevention	Vaccine, hygiene	Vaccine	—	—	Hygiene

2 Hepatitis B virus (HBV)

HBV can produce a wide range of outcomes: **asymptomatic carrier state** · **acute hepatitis with recovery** · **non-progressive chronic hepatitis** · **progressive chronic disease ending in cirrhosis** · **fulminant hepatitis** · **precursor to hepatocellular carcinoma**.

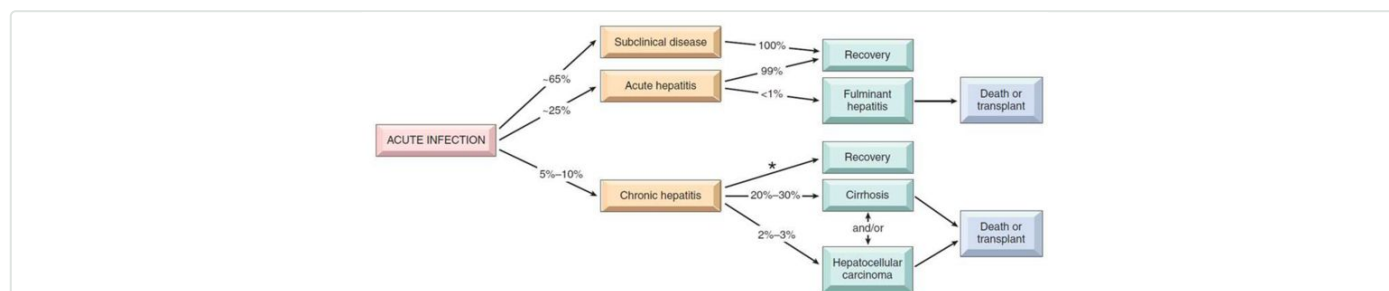
Interpreting hepatitis B serology

HBsAg	Anti-HBc	Anti-HBs	Interpretation
–	–	–	Susceptible (never infected)
–	IgG	IgG	Immune from natural infection
–	–	IgG	Immune from HBV immunization
+	IgM	–	Acutely infected
+	IgG	–	Chronically infected

HBsAg: positive in acute disease; continued presence = carrier state. **HBsAg**: indicator of transmissibility. **IgM anti-HBc**: marker of recent disease (window phase).

Outcomes of acute HBV infection

Pathway (from acute infection)	Outcome
~65% subclinical disease	100% recovery
~25% acute hepatitis	99% recovery · <1% fulminant hepatitis → death/transplant
5-10% chronic hepatitis	20-30% cirrhosis and/or 2-3% hepatocellular carcinoma → death/transplant (cirrhosis and HCC can also drive each other)



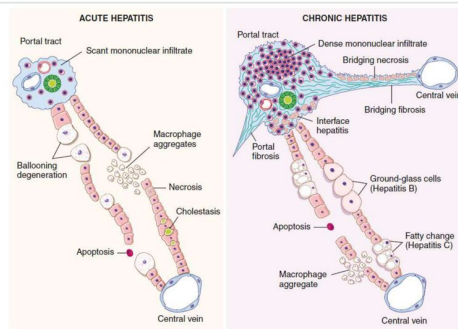
Potential outcomes of hepatitis B infection in adults

3 Hepatitis C virus (HCV)

Acute illness	Usually less severe than HBV.
Chronicity	Major cause of persistent infection/chronic hepatitis, developing in 80-90% of patients; cirrhosis in ~20% .
Transmission	Blood transfusion was the major route (now markedly reduced by screening); rising transmission via IV opioid drug abuse ; one-third have no identifiable risk factor.

4 Viral hepatitis — pathologic changes

Setting	Gross	Microscopic
Acute	Normal, enlarged (inflammation), or shrunken (massive necrosis/ acute liver failure)	Lobular disarray · hepatocyte swelling · apoptotic hepatocytes · portal & lobular lymphocytes · regeneration · cholestasis
Chronic	—	Defining feature: portal lymphocytic/lymphoplasmacytic inflammation with fibrosis . Chronic persistent : inflammation confined to portal tracts. Chronic active : inflammation spills into the parenchyma.



Acute vs chronic hepatitis: portal tract changes, interface hepatitis, bridging fibrosis/necrosis, ground-glass cells (HBV), fatty change (HCV)

5 Other liver infections

Routes of spread	Ascending (gut/biliary tract) · vascular seeding (portal system via GIT) · direct invasion (adjacent source, e.g. bacterial cholecystitis) · penetrating injury.
Examples	Bacteria (<i>Staph. aureus</i> , <i>Salmonella typhi</i> , <i>Treponema pallidum</i>) · <i>Entamoeba histolytica</i> (amebic liver abscess) · malaria · schistosomiasis · echinococcal infection (hydatid cyst) · liver flukes (predispose to cholangiocarcinoma in SE Asia).

6 Autoimmune hepatitis

What	Chronic, progressive disorder: genetic predisposition · association with other autoimmune disease · autoantibodies · responds to immunosuppression .
Presentation	Ranges from asymptomatic (elevated transaminases) to acute and chronic hepatitis.
Type	Antibodies
Type 1	ANAs · anti-smooth muscle antibodies (ASMA) · anti-SLA/LP
Type 2	Anti-liver kidney microsome-1 (anti-LKM-1) · anti-liver cytosol-1 (anti-LC1) · anti-SLA/LP

7 Drug- and toxin-induced liver injury

Why the liver	Major drug-metabolizing/detoxifying organ → exposed to many therapeutic and environmental chemicals; injury may be immediate or delayed by weeks-months.
Predictable reactions	Dose-dependent , e.g. acetaminophen toxicity.
Unpredictable (idiosyncratic) reactions	Depend on host-specific factors, e.g. chlorpromazine .

Pattern of injury	Causative agents
Fatty liver disease	Ethanol, steroids, aspirin
Cirrhosis	Ethanol, methotrexate, vitamin A
Hepatocellular adenoma	Oral contraceptives, anabolic steroids
Hepatocellular carcinoma	Alcohol, Thorotrast
Cholangiocarcinoma	Thorotrast
Angiosarcoma	Thorotrast, vinyl chloride
Cholestasis	Contraceptives, anabolic steroids, antibiotics, antiretrovirals
Cholestatic hepatitis	Antibiotics, phenothiazines, statins
Hepatocellular necrosis	Acetaminophen

P Must-remember pearls

- Hepatotrophic viruses: **HAV, HBV, HCV, HDV, HEV**. **HAV and HEV are fecal-oral, non-chronic** (HEV is severe in pregnancy); **HBV, HCV, HDV are parenteral/sexual and can become chronic**.
- **HBV serology**: HBsAg + IgM anti-HBc (no anti-HBs) = acute; HBsAg + IgG anti-HBc (no anti-HBs) = chronic; anti-HBs alone (IgG) = vaccine immunity; anti-HBc + anti-HBs (IgG) = natural-infection immunity.
- **Acute hepatitis** = scant portal infiltrate, ballooning degeneration, spotty necrosis. **Chronic hepatitis** = dense portal infiltrate with interface hepatitis, bridging fibrosis/necrosis; **ground-glass cells suggest HBV, fatty change suggests HCV**.
- **Autoimmune hepatitis type 1** = ANA/ASMA/anti-SLA-LP; **type 2** = anti-LKM-1/anti-LC1/anti-SLA-LP.
- **Predictable drug injury is dose-dependent (acetaminophen); idiosyncratic injury is host-dependent (chlorpromazine)**. Know the injury patterns: OCPs → adenoma; Thorotrast → HCC/cholangiocarcinoma/angiosarcoma; vinyl chloride → angiosarcoma.

Q Test yourself

1. A patient's serology shows HBsAg positive, IgM anti-HBc positive, and anti-HBs negative. This pattern indicates:

- A. Acute HBV infection
- B. Chronic HBV infection
- C. Immunity from vaccination
- D. Immunity from natural infection
- E. Susceptible, never infected

2. Liver biopsy in a patient with chronic hepatitis shows dense portal mononuclear infiltrate, interface hepatitis, and bridging fibrosis. Which additional finding would specifically suggest hepatitis B rather than hepatitis C as the cause?

- A. Fatty change
- B. Cholestasis
- C. Portal eosinophils
- D. Ground-glass hepatocytes
- E. Bile duct proliferation

3. A patient taking chlorpromazine develops liver injury unrelated to the dose taken, occurring in only a minority of patients on the drug. This is an example of:

- A. A predictable, dose-dependent reaction
- B. Autoimmune hepatitis
- C. An idiosyncratic (unpredictable) reaction
- D. A carrier state
- E. Fulminant viral hepatitis

4. Which autoantibody pattern is characteristic of type 2 autoimmune hepatitis?

- A. Antinuclear antibodies (ANA) and anti-smooth muscle antibodies (ASMA)
- B. Anti-mitochondrial antibodies
- C. Anti-HBs and anti-HBc
- D. Anti-dsDNA antibodies
- E. Anti-liver kidney microsome-1 (anti-LKM-1) and anti-liver cytosol-1 (anti-LC1)

5. A patient on long-term Thorotrast exposure is at increased risk for which combination of liver tumors?

- A. Hepatocellular adenoma only
- B. Hepatocellular carcinoma, cholangiocarcinoma, and angiosarcoma
- C. Hemangioma only
- D. Neuroendocrine tumor
- E. GIST

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — HBsAg+ with IgM anti-HBc and negative anti-HBs is the pattern of acute infection; IgG anti-HBc instead of IgM would indicate chronic infection.

2 • D — Ground-glass hepatocytes are a classic feature associated with chronic hepatitis B; fatty change is more associated with hepatitis C.

3 • C — Idiosyncratic reactions, like the response to chlorpromazine, depend on host-specific factors rather than dose, unlike predictable dose-dependent reactions such as acetaminophen toxicity.

4 • E — Type 2 autoimmune hepatitis is characterized by anti-LKM-1, anti-LC1, and anti-SLA/LP antibodies, whereas type 1 shows ANA, ASMA, and anti-SLA/LP.

5 • B — Thorotrast exposure is a classic cause of hepatocellular carcinoma, cholangiocarcinoma, and angiosarcoma of the liver.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL4 (titled “Liver, pancreas and biliary system” — slide content covers liver only), Dr Savan Saeed Azeez, Hawler Medical University
Study alongside the lecture, not instead of it.